

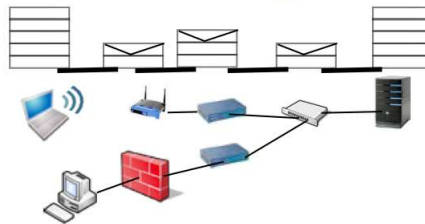
Computer Networking : Principles, Protocols and Practice, third edition

Contents

- [Computer Networking : Principles, Protocols and Practice, third edition](#)
- [Part 1: Principles](#)
- [Part 2: Protocols](#)
- [Part 3: Practice](#)
- [Appendices](#)

Computer Networking

Principles
Protocols
and
Practice



-
- [Preface](#)
 - [Connecting two hosts](#)
 - [The physical layer](#)
 - [The datalink layer](#)
 - [Framing](#)
 - [Recovering from transmission errors](#)
 - [Reliable data transfer on top of an imperfect link](#)
 - [Go-back-n and selective repeat](#)
 - [Building a network](#)
 - [The datagram organization](#)
 - [Computing forwarding tables](#)
 - [Flat or hierarchical addresses](#)
 - [Dealing with heterogeneous datalink layers](#)
 - [Virtual circuit organization](#)
 - [The control plane](#)
 - [Distance vector routing](#)
 - [Link state routing](#)
 - [Applications](#)
 - [The transport layer](#)
 - [Transport layer services](#)
 - [The connectionless service](#)
 - [The connection-oriented service](#)
 - [The request-response service](#)

- [Connection establishment](#)
 - [Data transfer](#)
 - [Connection release](#)
- [Naming and addressing](#)
 - [Benefits of names](#)
- [Sharing resources](#)
 - [Sharing bandwidth](#)
 - [Network congestion](#)
 - [Distributing the load across the network](#)
 - [Medium Access Control algorithms](#)
 - [Static allocation methods](#)
 - [ALOHA](#)
 - [Carrier Sense Multiple Access](#)
 - [Carrier Sense Multiple Access with Collision Detection](#)
 - [Carrier Sense Multiple Access with Collision Avoidance](#)
 - [Deterministic Medium Access Control algorithms](#)
 - [Congestion control](#)
 - [Congestion control with a window-based transport protocol](#)
- [The reference models](#)
 - [The five layers reference model](#)
 - [The Physical layer](#)
 - [The Datalink layer](#)
 - [The Network layer](#)
 - [The Transport layer](#)
 - [The Application layer](#)
 - [Reference models](#)
 - [The TCP/IP reference model](#)
 - [The OSI reference model](#)
- [Network security](#)
 - [Threats](#)
 - [Cryptographic primitives](#)
 - [Cryptographic protocols](#)
 - [Key exchange](#)
- [The application layer](#)
- [The Domain Name System](#)
- [Electronic mail](#)
 - [The Simple Mail Transfer Protocol](#)
 - [The Post Office Protocol](#)
- [The HyperText Transfer Protocol](#)
- [Making HTTP faster](#)
- [Remote Procedure Calls](#)
 - [Encoding data](#)
 - [Reaching the callee](#)
- [Remote login](#)
 - [The secure shell \(ssh\)](#)
- [Transport Layer Security](#)
 - [The TLS handshake](#)
 - [The TLS record protocol](#)
 - [Improving TLS](#)
- [Securing the Domain Name System](#)
- [Internet transport protocols](#)
- [The User Datagram Protocol](#)
- [The Transmission Control Protocol](#)
 - [TCP connection establishment](#)
 - [TCP reliable data transfer](#)

- [TCP's retransmission timeout](#)
 - [Advanced retransmission strategies](#)
 - [TCP connection release](#)
- [Congestion control](#)
 - [Controlling congestion without losing data](#)
 - [Modeling TCP congestion control](#)
- [The network layer](#)
 - [IP version 6](#)
 - [IPv6 addressing architecture](#)
 - [IPv6 packet format](#)
 - [ICMP version 6](#)
 - [The IPv6 subnet](#)
 - [Interactions between IPv6 and the datalink layer](#)
- [Routing in IP networks](#)
- [Intradomain routing](#)
 - [RIP](#)
 - [OSPF](#)
- [Interdomain routing](#)
 - [The Border Gateway Protocol](#)
 - [The BGP decision process](#)
 - [BGP convergence](#)
- [Datalink layer technologies](#)
 - [The Point-to-Point Protocol](#)
 - [Ethernet](#)
 - [Ethernet Switches](#)
 - [The Spanning Tree Protocol \(802.1d\)](#)
 - [Virtual LANs](#)
 - [802.11 wireless networks](#)
- [Exercises](#)
- [Reliable transfer](#)
 - [Open questions](#)
 - [Practice](#)
 - [Discussion questions](#)
- [Using sockets for inter-process communication](#)
 - [Sending data to a peer using a socket](#)
 - [Receiving data from a peer using a socket](#)
 - [connect: connecting a socket to a remote address](#)
 - [Creating a new socket to communicate through a network](#)
 - [Sending a message to a remote peer using its IPv6 address](#)
 - [Endianness: exchanging integers between different computers](#)
 - [Exercises](#)
- [Building a network](#)
 - [Multiple-choice questions](#)
 - [Open questions](#)
 - [Discussion questions](#)
- [Serving applications](#)
 - [Multiple choices questions](#)
 - [Open questions](#)
 - [Practice](#)
 - [Discussion questions](#)
- [Sharing resources](#)
 - [Medium Access Control](#)
 - [Fairness and congestion control](#)
 - [Discussion questions](#)
- [Application layer](#)

- [The HyperText Transfer Protocol](#)
- [TLS and ssh](#)
- [Analyzing packet traces](#)
- [TCP basics](#)
- [A closer look at TCP](#)
 - [Injecting segments in the Linux TCP stack](#)
 - [Open questions](#)
- [IPv6 Networks](#)
 - [Basic questions on IPv6 Networks](#)
 - [Design questions](#)
 - [Configuring IPv6 Networks](#)
 - [IPv6 packets](#)
- [Inter-domain routing](#)
 - [Exercises](#)
- [Exploring routing protocols](#)
 - [Exploring OSPF](#)
 - [Exploring RIP](#)
 - [Exploring BGP](#)
 - [Exercises](#)
- [Local Area Networks: The Spanning Tree Protocol and Virtual LANs](#)
 - [Exercises](#)
 - [Testing the Spanning Tree with IPMininet](#)
- [Glossary](#)
- [Bibliography](#)

Indices and tables

- [Index](#)
- [Search Page](#)



© Copyright 2019 Olivier Bonaventure.

Last updated on Nov 20, 2025.